Differential forms

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Wikipedia says that

> Differential forms provide a unified approach to define integrands over curves, surfaces, volumes, and higher-dimensional manifolds.

Let us try to figure out why that might be so.

Q. Given an open subset $U \subseteq \mathbb{R}^n$ and $1 \leq k \leq n$, what "structure" $X_{U,k}$ can I have on U such that for *every* k-manifold M inside U, there is a "nice" notion of taking "integral of $X_{U,k}$" over M?

A. Suppose that to every point $p \in U$, we have assigned a function φ_p which takes in k-parallelepipeds, and converts them into real numbers.

Now, given any k-manifold M, we can split it into many infinitesimal pieces which look like k-parallelepipeds, and then φ_p will tell us what value to assign to each parallelepiped. Finally, we take the sum over all of these values to compute the "integral" over M.

Note that we need φ_p to be "coherent" in assigning values to the k-parallelepipeds, for example, if I scale a parallelepiped A by 2 to get A', then $\varphi_p(A')$ should equal $2\varphi_p(A)$. This is so that the "integral" doesn't depend on which particular infinitesimal splitting of M we take.

This matches with the intuition we have developed in the section **Building intuition about exterior powers**. We want to have each $\varphi_p \in \Lambda^k(\mathbb{R}^n)^*$.

This "structure" that we have developed, is called a **differential k-form over U**.

Given an open subset $U \subseteq \mathbb{R}^n$, a **differential k-form over U** is a function ω of type

$$\omega :: (p : U) \to \left(\Lambda^k T_p(U)\right)^* \cong \Lambda^k T_p^*(U)$$

- We shall write "k-form" as short-hand for "differential k-form".
- Observe that 0-forms correspond to C^∞-functions on U.
- Just to summarize the intuition we have developed above,

> Defining a k-form on open subset $U \subseteq R^n$ is the same as assigning to each point in U its own "coherent" way of "measuring" k-parallelepipeds based at p.

Observe that every 1-form assigns linear functionals to every point $p \in U$.

When we take a derivative of a C^∞ function at a point in its domain, we get a linear functional.

Putting together the above two ideas, for every function $f \in C^\infty(U)$, we can define a 1-form df as

$$
\begin{aligned}
df &= Df \\
\text{i.e., } (df)_p &= Df(p) \\
\text{i.e., } (df)_p &= \left[v \in T_p(U) \mapsto D_v f(p) = Df(p)(v)\right]
\end{aligned}
$$

We have a basis $\{e_1, \ldots, e_n\}$ for $T_p(U) \cong \mathbb{R}^n$. Observe that

$$
\left(dx^i\right)_p\left(e_j\right) = D_j(x^i)(p) = \frac{\partial x^i}{\partial x_j} = \delta_{ij}
$$

Therefore, $\left\{\left(dx^i\right)_p\right\}$ is a basis for $T_p^*(U)$, and this basis is dual to $\{e_1, \ldots, e_n\}$, which is a basis for $T_p(U)$.

Note that here x^i is the "extract coordinate along e_i" function.

The 1-form dx over $\mathbb{R}^3$ assigns to each point $p \in \mathbb{R}^3$ the 1-volume-measuring-function given by

> For every 1-parallelepiped (i.e., a single vector) based at p, project it onto the X-axis, and check the extent of the shadow.

The 2-form $dx \wedge dy :: \left(p : \mathbb{R}^3\right) \to \Lambda^2\left(T_p^*\left(\mathbb{R}^3\right)\right)$ assigns to each point $p \in \mathbb{R}^3$ the 2-volume-measuring-function given by

> For every 2-parallelepiped based at p (i.e., a parallelogram, an element of $\Lambda^2 T_p(\mathbb{R}^3)$), calculate the (signed) area of its shadow on the XY-plane.

Given $f, g, h \in C^\infty\left(\mathbb{R}^3\right)$, the 2-form $f \cdot dx \wedge dy + g \cdot dy \wedge dz + h \cdot dx \wedge dz$ assigns to each point $p \in \mathbb{R}^3$ the 2-volume-measuring-function given by

> For every 2-parallelepiped based at p, calculate
>
> $$
> \begin{aligned}
> &f(p) \cdot (\text{signed area of shadow on XY-plane}) + \\
> &g(p) \cdot (\text{signed area of shadow on YZ-plane}) + \\
> &h(p) \cdot (\text{signed area of shadow on XZ-plane})
> \end{aligned}
> $$

Note that when we say "signed area of shadow on XY-plane", we refer to the "oriented" XY-plane. So, "signed area of shadow on YX-plane" will be $(-1) \cdot$ "signed area of shadow on XY-plane".
